

Pathogen Outbreak Tabletop Exercise — Unified Executive Summary Dossier

Multi-AI Synthesis Report

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APPROVAL NOTICE: This document is pending electronic approval and ONESTAMP from the author. No elements herein should be treated as final, approved exercise design until ONESTAMP is applied. All recommendations, frameworks, open questions, and action items are advisory until that approval is granted. This document is a synthesis and analysis of contributed AI responses — it is a planning instrument, not an operational directive.

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1. Why This Exercise Matters

The Preparedness Imperative

Tabletop exercises (TTXs) exist because the worst time to discover a gap in your response system is during an actual emergency. For pathogen outbreaks specifically, the critical failure window is not the declared pandemic phase — it is the pre-recognition window: the hours and days before anyone knows what they are dealing with, before protocols exist, before supply chains have been activated, and before guidance has arrived from any authority.

History has demonstrated this pattern repeatedly:

- **SARS (2003):** Healthcare workers in Hong Kong and Toronto were infected and dying before the pathogen was identified or transmission routes were understood. The failure point was not the response — it was the gap between first presentation and recognition.
- **H1N1 (2009):** Initial cases were misidentified and mishandled. The first critical days involved decisions made without diagnostic certainty.
- **COVID-19 (2019–2020):** Emergency departments across the United States saw patients with an unknown respiratory illness for weeks before any protocol, PPE guidance, or diagnostic tool existed.
- **Ebola (2014–2016):** A single unrecognized case arriving in a U.S. hospital revealed critical gaps in isolation protocol, staff communication, and PPE training that the system believed it had covered.

In every case, the systems that failed were not tested before they were needed. The systems that performed were the ones whose people had rehearsed ambiguity — not just protocols.

Why Now

Major outbreak events do not follow a fixed schedule, but several converging risk factors make the present a heightened-risk window:

- Post-COVID global travel density has returned to pre-pandemic levels while surveillance infrastructure in many regions remains degraded
- Zoonotic spillover pressure is increasing due to habitat encroachment, climate-driven animal migration shifts, and agricultural density

- Antimicrobial resistance is eroding the safety net that previously contained secondary bacterial infections during viral outbreaks
- Institutional memory from COVID-19 is beginning to fade — personnel have turned over, emergency protocols have been archived, and supply chain lessons have not been systematically embedded
- AI-accelerated biomedical research has dramatically shortened the timeline between pathogen discovery and broader dissemination of synthetic biology knowledge

The question is not whether another major outbreak will occur. The question is whether the people and systems responsible for response will have rehearsed what to do when they do not know what to do.

This tabletop exercise is designed to answer that question before it becomes urgent.

2. Executive Summary

The **Pathogen Outbreak Tabletop Exercise (TTX)** is a preparedness planning initiative designed to stress-test the earliest, most ambiguous phase of a novel pathogen outbreak — the pre-recognition window where no protocol exists, no supplies are staged, and clinical decision-making operates entirely on provider judgment under uncertainty.

The concept was submitted to five AI systems — ChatGPT, Google Gemini, Microsoft Copilot, Grok AI, and Claude AI — for independent analysis, design recommendations, and framework contributions. This document synthesizes all five responses into a unified, actionable reference dossier. Perplexity Computer serves as the compiling and contributing sixth intelligence for this document.

Five-AI Consensus Summary

Topic	Cross-AI Consensus
Concept strength	All 5 AIs rate the concept as strong and professionally grounded — comparable to FEMA, CDC, and WHO TTX frameworks
Defining innovation	All 5 AIs identify the “residency hook” (provider daily uncertainty as the exercise baseline) as the single most powerful and differentiating design element
Most critical gap	All 5 AIs agree: no pathogen archetype defined — everything downstream is blocked until this decision is made
Second most critical gap	All 5 AIs agree: no primary objective locked — exercise will produce unfocused findings without one

Topic	Cross-AI Consensus
"Every 10 years" framing	All 5 AIs flag this as a credibility risk with sophisticated participants; recommend reframing as a probability statement, not a schedule
Equity layer	All 5 AIs (led by Claude and Grok) identify missing equity/differential impact analysis as a significant blind spot
Highest design risk	All 5 AIs flag "Nathan as omnipotent architect" / facilitator bias risk — the exercise must be designed to reveal failure, not confirm competence
Recommended scale	Majority (4 of 5 AIs) recommend starting at single or dual hospital + county public health — deep over broad

3. The Core Concept — What Was Proposed

The submitted concept had two components:

Component 1 — The Strategic Framework

A tabletop exercise designed around a pathogen outbreak, built on the following pillars:

- **Purpose:** To establish and test response capabilities before the next major outbreak event
- **Justification:** Historical outbreak cadence suggests a heightened-risk window
- **Scope and scale:** To be determined
- **Participants:** To be determined
- **What is being tested:** Response systems, coordination, and decision-making under uncertainty

This reflects the correct structural approach used by professional TTX designers operating under CDC, WHO, FEMA/HSEEP, and RAND frameworks.

Component 2 — The Provider Uncertainty Hook

A second, critically important framing element was introduced for medical professionals:

"When you did your residency, how many times did a patient present to you and you didn't know what to do? Every day? That will be every day of this scenario in the beginning. It will not be predictable, it will not be something you can prepare for, it will not be something you have supplies for. That is where we will begin."

This framing is unanimously recognized across all five AIs as the defining innovation of the concept. It:

- Bypasses institutional defensiveness (“we already have a plan for that”)
 - Anchors the exercise in lived clinical experience rather than theoretical protocol
 - Establishes maximum ambiguity as the baseline condition — not an escalation state
 - Transforms the exercise from an “outbreak simulation” into an “uncertainty-driven system stress test”
 - Directly targets the pre-recognition phase that most exercises skip entirely
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4. What Works — Strengths Across All AI Contributions

The following strengths are validated across all five AI responses:

4.1 Structural Soundness

The concept is already built on the correct TTX pillars (purpose, testing parameters, relevance, scope, participants). This is the exact framework used by FEMA, CDC, and WHO in professional-grade exercises. No redesign is needed at the structural level — only completion.

4.2 The Residency Hook — Clinical Uncertainty as Exercise Baseline

This is not a rhetorical device. It is a precision exercise design choice that does several things simultaneously:

- **Psychological buy-in:** Providers recognize the feeling immediately and engage rather than defend
- **Realism:** It accurately represents the actual failure point of historical outbreaks
- **Scope definition:** It anchors the exercise squarely in the pre-recognition phase
- **Differentiator:** Most institutional TTXs test the response phase; this one tests the detection and decision phase

Claude identifies this as “the single most powerful design choice” and the element that “bypasses the institutional defensiveness that kills most tabletop exercises.”

4.3 System-Level Thinking

The concept is not testing a disease — it is testing how interconnected systems behave when their assumptions collapse. This is a fundamentally more sophisticated and more valuable framing. Copilot notes that this tests “cross-sector coordination: public health, clinical care, labs, logistics, communications” in a single scenario.

4.4 Proactive Timing

Designing this exercise before an outbreak rather than after places participating institutions ahead of the preparedness curve. Post-COVID institutional memory is still active — participants will bring real emotional and operational experience to the discussion.

4.5 Scalability

The core concept works at hospital level, county level, state level, or national level without requiring fundamental redesign. The provider uncertainty layer remains valid at every scale.

4.6 Low Cost, High Yield

A TTX reveals policy, communication, and coordination gaps without the expense of a full functional exercise. It builds relationships, tests plans without real-world risk, and surfaces surprises in areas where institutions believe they are prepared.

5. What Doesn't Work (Yet) — Gaps and Risks

The following gaps are flagged across multiple AI responses. None are fatal — all are solvable with author decisions.

5.1 No Pathogen Archetype Defined — CRITICAL

This is the single largest structural gap. Every downstream design element — injects, decision points, resource stress conditions, evaluation metrics, participant roles — depends on knowing the pathogen characteristics. Transmission mode, incubation period, severity spectrum, and diagnostic detectability determine the entire exercise. **Nothing can be finalized until this decision is made.**

5.2 No Primary Objective Locked — CRITICAL

Multiple possible objectives are embedded in the concept: clinical decision-making, inter-agency coordination, supply chain resilience, public communication, provider psychological load. An exercise without a single primary objective produces unfocused findings that don't translate into funded improvement actions. One sentence. Author decision required.

5.3 The “Every Ten Years” Framing — CREDIBILITY RISK

As stated, this implies that pathogens follow a calendar — which they do not. Epidemiologists, public health officers, and federal stakeholders will challenge this immediately, and it risks derailing early exercise framing. **Recommended replacement (cross-AI consensus):**

“Major outbreak events have occurred with increasing frequency and unpredictability, and current global conditions — including zoonotic spillover pressure, antimicrobial resistance, and global travel density — represent a heightened-risk window that justifies proactive exercise now.”

5.4 Participant Roles Not Structured

“Medical professionals” is not a participant structure. The exercise needs defined role categories with clear decision authority mapped to each: who can commit resources, who escalates, who communicates externally, who documents. Without this, the exercise cannot simulate real interaction.

5.5 No Evaluation Architecture

What does success look like? What does failure look like? Without pre-defined metrics, the after-action report will be anecdotal rather than actionable. This needs to be built into the design, not added afterward.

5.6 Chaos Must Be Bounded

“Everything is unknown all the time” does not work in an exercise — participants will freeze or disengage. The exercise needs layered uncertainty, not total randomness. Unknown symptoms should evolve into emerging patterns; conflicting guidance should resolve into partial clarity over time. The goal is controlled disorientation, not paralysis.

5.7 “You Can’t Prepare” Framing Needs Calibration

The residency hook is powerful, but if delivered without careful framing, “you can’t prepare for this” risks inducing participant shutdown rather than productive engagement. The exercise must communicate: *you cannot prepare with protocols, but you can prepare with frameworks for thinking under uncertainty*. That distinction is critical.

5.8 Missing Equity Layer

No outbreak scenario in a major metropolitan area is complete without examining differential impact on under-resourced communities, non-English-speaking populations, unhoused individuals, and long-term care residents. Omitting this produces a TTX that reflects the best-case version of the healthcare system, not the actual one.

5.9 No Evaluation Baseline Established

If this is the first exercise of its kind for participating institutions, there is no baseline against which to measure performance. Findings will be descriptive but not benchmarkable. The first exercise should explicitly establish that baseline so future exercises can show trajectory.

5.10 Stakeholder Buy-In Not Secured

If key partners (ED leadership, Infection Prevention, hospital administration, public health liaison) are not committed before design work proceeds, the exercise becomes a paper exercise with limited operational value.

6. What Needs to Be Addressed Before Moving Forward

Immediate Author Decisions (Block All Downstream Work)

#	Decision Required	Why It Blocks
D1	Lock the pathogen archetype	Every inject, decision point, resource stress, and evaluation metric depends on pathogen characteristics
D2	Lock the primary objective (one sentence)	Exercise will produce unfocused findings without a single measurable primary goal
D3	Define target audience and scale	Participant structure, inject depth, and evaluation scope all depend on this
D4	Establish sponsoring organization and authority structure	Who convenes the exercise, who can commit resources during play, who owns the AAR
D5	Define evaluation baseline	Is this a first-time exercise for participants? Must be documented to frame findings correctly

Design Work Required Before Exercise Date

1. Draft and review Situation Manual (SitMan)
2. Build facilitator script with probe questions for each inject
3. Establish ground rules document (no-fault, confidentiality, decision authority limits)
4. Develop participant roles and pre-read package
5. Build evaluation forms and note-taking templates
6. Confirm clinical Subject Matter Expert (SME) to adjudicate injects in real time
7. Establish media simulation protocol for Phase 2 inject
8. Draft hotwash and After-Action Report (AAR) templates before exercise — not after
9. Secure preliminary buy-in from ED leadership, Infection Prevention, Lab, and Hospital Administration
10. Define legal and privacy constraints that may limit realistic decision-making during the TTX

7. Critical Open Questions Registry

All questions below require author resolution before design can advance. HIGH priority items block Phase 1 design work.

Category 1 — Scenario & Pathogen

#	Question	Options	Priority
Q1	What type of pathogen features in this scenario?	Novel respiratory virus (SARS-like) / Unknown zoonotic / Vector-borne / Bacterial / Hemorrhagic fever / “Disease X” — uncharacterized	HIGH
Q2	Is the pathogen natural or potentially engineered?	Natural spillover / Suspected engineered / Ambiguous (never confirmed)	HIGH
Q3	What is the transmission mode?	Airborne / Droplet / Contact / Vector / Waterborne / Multi-modal	HIGH
Q4	What is the severity profile?	High transmissibility, moderate mortality / Low transmissibility, high mortality / Variable / Primarily affects specific populations	HIGH
Q5	What is the incubation period?	24–48 hours (fast) / 2–10 days (moderate) / 2–3 weeks (slow)	HIGH
Q6	Do standard rapid diagnostics (flu, COVID, RSV) identify it?	Yes / No (negative result) / Inconclusive / Test-dependent	HIGH

Category 2 — Scope & Scale

#	Question	Options	Priority
Q7	What is the geographic scope?	Single hospital / Multi-hospital system /	HIGH

#	Question	Options	Priority
		County-level (e.g., Bakersfield / Kern County) / Regional (multi-county) / State (California) / National	
Q8	Which organization's response is primarily being tested?	Hospital clinical operations / County public health / Hospital + county public health (joint) / Multi-agency coalition	HIGH
Q9	Should this TTX include federal agency simulation?	Yes (CDC, HHS, FEMA as simulators) / No (local/state only) / Optional inject	MEDIUM
Q10	What is the intended exercise duration?	Half-day (4 hours) / Full day (8 hours) / Multi-day / Series of shorter sessions	MEDIUM

Category 3 — Objectives & Testing

#	Question	Options	Priority
Q11	What is the single primary objective (one sentence)?	Author to define — see Section 8.1 for recommended draft	HIGH
Q12	What is the highest-priority capability to test?	Clinical detection and escalation / PPE and supply chain management / Inter-agency coordination / Public communication / All four (requires multi-day exercise)	HIGH
Q13	What specific decisions must participants be able to make during the scenario?	Author to define — see Section 8.4 for recommended decision points	HIGH
Q14	What does "success" look like in measurable	Author to define — see Section 8.6 for	HIGH

#	Question	Options	Priority
	terms?	recommended metrics framework	

Category 4 — Clinical Design

#	Question	Options	Priority
Q15	What symptoms do the first patients present with?	Author to define (general characteristics; no real pathogen protocols required)	HIGH
Q16	What is the initial misdiagnosis or diagnostic dead end?	Flu (negative) / COVID (negative) / Atypical pneumonia / Undifferentiated systemic illness	HIGH
Q17	How quickly does clinical deterioration occur in early cases?	Unexpectedly rapid / Standard / Variable and misleading	HIGH
Q18	Is there a pediatric or geriatric population specifically affected?	Yes (shapes triage decisions) / No / Disproportionately but not exclusively	MEDIUM

Category 5 — Authority, Ethics & Legal

#	Question	Options	Priority
Q19	Is the Architect / Facilitator authorized to let failure occur on record?	Yes (required for valid findings) / Partially (some limits) / Author to define limits	HIGH
Q20	What legal and privacy constraints limit realistic decision-making during the TTX?	Author to define before exercise design proceeds	HIGH
Q21	Who owns the After-Action Improvement Plan and how will it be funded?	Named organization / Named individual / Author to assign	HIGH
Q22	Will the TTX include	Yes (required) / Scope	HIGH

#	Question	Options	Priority
	explicit no-fault and confidentiality ground rules?	to be defined by author	

8. Recommended Exercise Architecture

The following architecture is synthesized from all five AI contributions. All elements are DRAFT — pending author decisions from Section 7.

8.1 Purpose & Primary Objective

Primary Purpose (recommended — pending author approval): To stress-test the earliest, most ambiguous phase of a novel pathogen outbreak — specifically the pre-recognition window where no protocol exists, no supplies are staged, and clinical decision-making operates entirely on provider judgment under uncertainty.

Primary Objective (recommended — pending author approval): Evaluate how frontline clinical teams detect, escalate, and make containment decisions during the first 72 hours of an uncharacterized pathogen outbreak, before any formal guidance or diagnostic confirmation exists.

Supporting Objectives:

- Test internal hospital communication pathways under diagnostic uncertainty
- Evaluate PPE and isolation resource decision-making under scarcity conditions
- Identify escalation thresholds — at what point do providers alert Infection Prevention, administration, and public health
- Assess provider cognitive and operational load during sustained uncertainty
- Evaluate inter-agency handoff protocols between hospital and public health systems
- Test communication and public messaging preparedness before institutional guidance is ready

8.2 Recommended Pathogen Archetype

(Pending author selection from Q1–Q6. The following is the cross-AI recommended default.)

A novel zoonotic respiratory pathogen with the following working characteristics:

Characteristic	Working Value
Transmission	Airborne and droplet — moderate-to-high transmissibility
Incubation period	Variable: 2–10 days — defeats early contact tracing

Characteristic	Working Value
Initial symptoms	Non-specific: fever, myalgia, atypical respiratory symptoms, occasional dermatological manifestations
Rapid diagnostic result	Negative on standard flu, COVID, RSV panels
Severity spectrum	Moderate-to-high severity in 15–20% of cases; mild or asymptomatic in remainder — creates misleading early mortality signal
Treatment availability	No established protocol; no available vaccine
Lab confirmation timeline	48–72 hour turnaround for specialized testing

Why this archetype (cross-AI rationale): It maximizes the provider uncertainty layer, defeats standard diagnostic assumptions, creates realistic PPE burn-through before recognition, produces a delayed and confusing public health signal, and stresses exactly the systems this exercise intends to test. It is also the archetype closest to the conditions that produced the largest real-world response failures (SARS, early COVID-19).

Alternative archetypes for author consideration:

- High transmissibility / lower mortality respiratory (influenza-like) — tests surge capacity
- Low transmissibility / high mortality hemorrhagic (Ebola-like) — tests isolation and staff safety protocols
- Vector-borne with atypical presentation — tests geographic mapping and contact investigation
- Unknown bacterial with antibiotic resistance — tests empiric treatment decision-making

8.3 Participant Structure

Primary Players (active decision-makers):

- Emergency Department physicians and charge nurses
- Hospitalists and ICU attending physicians
- Infection Prevention and Control officers
- Hospital administration (CMO, CNO, COO)
- Laboratory directors
- Pharmacy leads
- Respiratory therapy leads
- Supply chain / materials management

Secondary Players (consulted and reactive):

- Public health liaison (county or state level)
- Legal counsel

- Risk management
- Communications / public affairs

Evaluators and Observers:

- Clinical Subject Matter Expert (adjudicates inject realism in real time)
- Exercise facilitator
- Note takers assigned to each breakout group
- Ethics representative

Simulators (played by control team):

- Off-site hospitals reporting similar cases
 - State and federal public health agencies
 - Media inquiries
 - Specialty consultants
 - Supply vendors
-

8.4 Exercise Timeline & Master Scenario Events List (MSEL)

Phase 1: The Anomaly — “Every Day is the First Day” (Hours 0–48)

Opening Facilitator Frame: *“You are in Week 1. A patient presents. Rapid panels are negative. You don’t know what you’re looking at. When you did your residency, how many times did a patient present to you and you genuinely didn’t know what to do? Every day? That is today — except now it’s happening with multiple patients per shift, and you have no playbook, no surge supplies, and no clear guidance yet. That is where we begin.”*

Inject	Event
Inject 1	First patient presents — 38-year-old, fever, myalgia, atypical respiratory symptoms, all rapid panels negative. No obvious exposure history. Standard workup unrevealing.
Inject 2	Second patient at a different facility, same metropolitan area — similar presentation. Labs inconclusive. Both patients deteriorate faster than expected.
Inject 3	Nursing staff expresses discomfort with standard precautions. Two nurses request N95s — supply room has limited stock and no authorization pathway.
Inject 4	Infection Prevention is notified. No protocol

Inject	Event
	match exists. They begin drafting ad hoc guidance with incomplete information.
Inject 5	Hospital administration requests a status update. Providers cannot give a clear answer.

Decision Points — Phase 1:

- Do you isolate? Under what category? In what physical space?
- Do you report to public health? At what threshold — suspicion, probability, or confirmed case?
- Do you begin empiric treatment? With what, and who authorizes it?
- How do you communicate diagnostic uncertainty to patients and families?
- Who authorizes elevated PPE use before any pathogen confirmation?

Phase 2: Pattern Recognition & System Strain (Hours 48–120)

Inject	Event
Inject 6	Third and fourth cases confirmed at two additional facilities. State public health begins inquiry.
Inject 7	PPE supplies at 60% of normal capacity and burning fast due to precautionary use across all suspected cases.
Inject 8	One provider develops symptoms and is furloughed. Staffing pressure begins.
Inject 9	Lab confirms novel pathogen — unknown classification. CDC notified. 48-hour wait for further guidance.
Inject 10	Local news outlet reports “mysterious illness” at area hospitals. Phone lines begin lighting up with calls from the public, families, and other providers.
Inject 11	Supply chain vendor notifies hospital that N95 reorder will be delayed 7–10 business days.

Decision Points — Phase 2:

- How do you manage the furloughed staff member and potentially exposed colleagues?
- How do you respond to media inquiry before official guidance exists?
- How do you triage limited PPE across departments with insufficient stock?
- What do you tell the public health agency when you still don’t have answers?

- How do you maintain normal hospital operations while managing suspected cases?
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Phase 3: Coordinated Response & Transfer (Hours 120–168)

Inject	Event
Inject 12	CDC issues preliminary guidance — partially inconsistent with ad hoc protocols already in place.
Inject 13	Surge begins. Alternate care site activation is discussed by hospital leadership.
Inject 14	A long-term care facility reports a cluster. Hospital receives a transfer request for multiple residents.
Inject 15	Ethics consult is requested — limited ventilator availability and two critical patients with equal clinical priority.
Inject 16	Hospital is asked to participate in a regional coordination call with 12 other facilities.

Decision Points — Phase 3:

- How do you reconcile external CDC guidance with decisions and protocols already implemented?
 - What are your surge activation thresholds and alternate care site criteria?
 - How do you handle the ethics consult under explicit resource scarcity?
 - Who speaks for your institution on the regional coordination call?
 - What does your after-action documentation look like at this stage?
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8.5 Constraints & Stress Conditions

The following constraints are active throughout the exercise and are not negotiable within the scenario:

- No established diagnostic protocol for the first 72 hours of exercise time
- Standard rapid diagnostic panels return negative results throughout Phase 1
- PPE supply is finite and non-replenishable during the exercise window
- External guidance is delayed, evolving, and occasionally contradictory
- Staffing is degraded by provider exposure and precautionary furlough beginning Phase 2
- Media pressure begins before institutional messaging is prepared
- Resource allocation decisions must be made before an ethics framework is formally invoked
- All decisions are time-pressured — facilitator will impose decision windows per inject

- Information released to participants is asymmetric — not all players have the same data
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8.6 Evaluation Criteria & Metrics

Time Metrics:

- Time from first presentation to isolation decision
- Time from first presentation to Infection Prevention notification
- Time from Infection Prevention notification to public health notification
- Time to activate surge protocols

Decision Quality Metrics:

- Was the isolation decision defensible given available information?
- Was the escalation threshold appropriate and consistently applied?
- Was PPE rationing documented and justified?
- Was internal provider communication consistent across departments?

Resource Metrics:

- PPE utilization rate versus available supply
- Isolation bed utilization and capacity management
- Time to request external supply support

Communication Metrics:

- Clarity and consistency of internal communications across departments
- Quality and timing of public health notification
- Preparedness of external messaging before media contact

Human Factors Metrics:

- Identified decision-making bottlenecks and where they occurred
 - Reported cognitive overload points among participants
 - Role clarity under sustained pressure
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8.7 After-Action Framework

Immediate Hotwash (30 minutes post-exercise):

- What went well?
- What surprised you?
- What would you do differently?

After-Action Report (delivered within 2 weeks):

- Findings mapped to each primary and supporting objective
- Gaps identified with evidence from exercise record
- Recommendations with assigned owners
- Priority ranking: Immediate / 90-Day / Long-Term

Improvement Plan:

- Each recommendation tied to a named owner, a resource requirement, and a follow-up exercise or drill date
 - Tracked quarterly
 - AAR submitted for ONESTAMP ratification before integration into institutional improvement record
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9. Red-Team Risk Analysis

The following risks are synthesized from all five AI red-team contributions. Each risk includes severity, description, and mitigation.

9.1 Technical & Design Risks

Risk	Description	Severity	Mitigation
Facilitator Bias — The Architect Problem	If the exercise is facilitated by someone too invested in a particular outcome, injects will unconsciously confirm rather than stress. Real failure modes will be avoided.	HIGH	The facilitator must be genuinely willing to let the exercise reveal failure — including failure that reflects poorly on existing plans or leadership. Consider an external facilitator or co-facilitation structure.
The Protocol Comfort Reflex	Experienced participants will instinctively reach for existing protocols even when the scenario explicitly renders them inapplicable.	HIGH	Injects must be constructed to make protocol-dependency visible as a gap, not a solution. The facilitator must actively redirect protocol-reach back to the scenario conditions.

Risk	Description	Severity	Mitigation
Scope Creep During Execution	TTXs reliably tend to expand mid-exercise as participants follow interesting threads. Phase 1 discussions will consume the entire exercise and Phase 3 will never occur without discipline.	HIGH	Hard time boundaries per phase; facilitator authority to cut discussion; pre-set inject triggers on a clock regardless of where discussion stands.
The “No Wrong Answer” Trap	TTX designs can be so no-fault that participants never feel genuine pressure. Without pressure, the residency hook loses its power.	HIGH	Build controlled stress conditions into the design; facilitator must apply time pressure to decision points; some decisions must be made with incomplete information, on the record.
Evaluation Without Baseline	Without a baseline, findings are descriptive but not benchmarkable. You cannot measure improvement without a starting point.	MEDIUM	Document explicitly that this is a baseline-establishing exercise; create a standardized scoring rubric that future exercises can reuse.
Omnipotent Instant Resolution	If participants can always access perfect information by asking the right person, the exercise collapses.	MEDIUM	Information asymmetry must be scripted and enforced — different players receive different data; some requests for information must return “unavailable” or “delayed.”

9.2 Narrative & Operational Risks

Risk	Description	Severity	Mitigation
Provider Alienation if Opening is Mishandled	The residency hook is powerful but needs careful delivery. If it	HIGH	The hook must land as: “This is the real condition we are

Risk	Description	Severity	Mitigation
	reads as “you will fail and there’s nothing you can do,” it creates defensive resistance rather than engagement.		designing for, and your experience navigating uncertainty is exactly what this exercise is built around.” Tone and delivery of the opening are as important as content.
Missing the Equity Layer	No outbreak scenario in a major metropolitan area is complete without examining differential impact on under-resourced communities, non-English-speaking populations, unhoused individuals, and long-term care residents. Omitting this produces a TTX that reflects a best-case healthcare system, not the actual one.	HIGH	Introduce at least two injects that force participants to make decisions about populations with additional access barriers; evaluate response for equity gaps in the AAR.
Legal and Liability Paralysis	Without pre-established no-fault ground rules and confidentiality protections, legal counsel and risk management participants will significantly constrain discussion quality.	HIGH	Ground rules must be explicit and agreed upon — ideally signed — before the exercise begins. Scope what can and cannot be said on record.
Repetition of COVID Narrative	Participants with strong COVID-19 institutional memory may unconsciously default to COVID-	MEDIUM	The pathogen must be clearly distinguished from COVID-19 in both presentation and clinical profile;

Risk	Description	Severity	Mitigation
	specific responses rather than engaging with the novel pathogen scenario.		facilitator must redirect COVID-specific protocol references back to the scenario's unknown conditions.
Overloading with Simultaneous Systems Tests	Testing clinical operations, public communications, supply chain, and inter-agency coordination simultaneously in one exercise will dilute learning across all areas.	MEDIUM	Define a primary focus clearly; structure secondary objectives as depth-testing moments rather than parallel exercise tracks.

9.3 Recommended Minimum Requirements Before Launch

The following must be in place before the exercise is considered ready to run:

1. At least one canonical failure mode is built into the design — the exercise must have a designed failure arc
2. Ground rules document is completed and signed before exercise begins
3. Clinical SME is confirmed to adjudicate injects in real time
4. AAR template is drafted before the exercise, not after
5. Equity layer is incorporated into at least Phase 2 or Phase 3 injects
6. Pathogen archetype is locked (D1) — no exercise can begin without this

10. Per-AI Contribution Breakdown

This section documents what each contributing AI added, what was uniquely theirs, and where their perspective diverged from or reinforced the others.

ChatGPT — Structural Framework & Clinical Reality Model

Round 1 Contribution: ChatGPT provided the initial structural validation and the most comprehensive breakdown of the concept's components. It confirmed that the proposed concept matches professional TTX frameworks (FEMA, CDC, DHS) and produced the clearest articulation of what is missing: no defined objectives, no scenario constraints, no participant structure, and no evaluation criteria. ChatGPT introduced the formal "Inject #1" framing and suggested a 9-section exercise structure.

Round 2 Contribution: After the provider uncertainty hook was introduced, ChatGPT formalized it into a “Clinical Reality Framework” layer. This was the first AI to fully operationalize the residency analogy into a reusable exercise design element — naming it the “uncertainty-driven system stress test” framing. ChatGPT also introduced the concept of layered uncertainty (as opposed to total chaos) and the requirement for provider action constraints (“what they are allowed to do when they don’t know”).

Unique to ChatGPT:

- First to call this an “uncertainty-driven system stress test” (not an “outbreak simulation”)
- Introduced the 4-phase progression model: Ambiguity → Pattern Recognition → System Strain → Coordinated Response
- Clearest articulation of the requirement for specific measurable objectives

Divergence: ChatGPT was the most focused on the provider-level clinical design and least focused on the multi-agency coordination and legal/ethical dimensions that Copilot and Claude developed more fully.

Google Gemini — Geographic Context, MSEL Architecture & Phase Design

Contribution: Gemini brought the most detailed Master Scenario Events List (MSEL) architecture and was the first to specifically anchor the exercise in a recognizable geographic and institutional context (Bakersfield / Los Angeles County, LACDPH, California Department of Public Health). Gemini also introduced the “Disease X” / novel uncharacterized pathogen framing and proposed the “Resource Trap” as a designed exercise element — PPE burn-through before recognition as an intentional stressor.

Gemini’s Sample Inject (strongest in the document):

“You’re in Week 1. A 38-year-old patient presents to your ED with fever, cough, and fatigue. Rapid flu/RSV/COVID panels are negative. Chest X-ray is nonspecific. The patient works at LAX and recently traveled domestically. When you did your residency, how many times did a patient present to you and you genuinely didn’t know what to do? Every day? That is today — except now it’s happening with multiple patients per shift, and you have no playbook, no surge supplies, and no clear guidance yet. Walk us through your first 30 minutes with this patient and your team.”

Unique to Gemini:

- Specific geographic anchoring (Bakersfield, LA County, LACDPH)
- “The Resource Trap” as a designed structural element
- Three-module TTX sketch (Detection/Chaos, Escalation, Decision Points/Recovery)
- Proposed one-page purpose statement as a model for author approval
- Clearest proposed scope: 2–3 hospitals + county public health as the optimal starting configuration

Divergence: Gemini was the most location-specific of the five AIs, which provides the most actionable design but also the most assumptions about context that may need adjustment.

Microsoft Copilot — Policy Architecture, Legal/Ethics & Evaluation Framework

Contribution: Copilot produced the most structured policy and logistics framework of any contributor. Its “High Level Appraisal” table format and detailed checklist approach brought the clearest pre-exercise planning scaffolding. Copilot was the first to systematically address legal, ethical, and privacy considerations, and the only AI to explicitly raise the question of stakeholder buy-in as a design prerequisite.

Copilot’s Tier-Fidelity framework from the previous document (see OAL-DOSSIER-CANON-002) was adapted here into a clinical context: which patient populations and systems require the highest-fidelity simulation, and which can receive lower-resolution design without compromising exercise validity.

Unique to Copilot:

- Most comprehensive legal and ethical coverage (data sharing, patient privacy, quarantine authority, liability)
- Only AI to explicitly flag the budget and sponsorship question
- Introduced the “Players vs. Observers vs. Evaluators” distinction with formal evaluation team structure
- Provider-focused clinical objectives translated into measurable time metrics
- Reference to ASPR TRACIE and pediatric pandemic network toolkits for accelerating materials development

Divergence: Copilot was the most process-oriented and logistics-focused, which is essential but also the most institutionally conservative — it was the least willing to push into novel or aggressive design territory.

Grok AI — Equity, Local Context & Provider Psychological Realism

Contribution: Grok brought the strongest community and equity framing, and the most emotionally intelligent analysis of the provider uncertainty hook. Grok was the first to name the “emotional and experiential resonance” dimension of the residency framing explicitly and to argue that it “lowers defenses and encourages honest discussion instead of ‘we already have a plan for that.’”

Grok also provided the most detailed geographic and demographic context for the Los Angeles setting — specifically flagging homelessness, international airport exposure pathways, dense and diverse populations, and vulnerable communities as exercise design variables.

Unique to Grok:

- Most explicit equity and community impact framing
- Introduced the LACDPH / LA County emergency management context most thoroughly
- Raised bioterrorism sensitivity as a potential constraint on exercise scope
- Most detailed key questions registry (9 questions with structured option sets)
- Proposed the most specific sample purpose statement (ready to adopt or modify)
- Called out the importance of existing plans/policies as the testing baseline

Divergence: Grok was the most focused on the Los Angeles-specific context; some elements will require adjustment if the exercise is conducted in a different jurisdiction.

Claude AI — Full Exercise Architecture, Red-Team Analysis & Canonical Integration

Contribution: Claude provided the most comprehensive and structurally complete response of all five AIs — including a full 8-section exercise outline, a detailed MSEL with 16 injects across 3 phases, specific decision points for each phase, a full evaluation criteria framework, and the most rigorous red-team analysis. Claude also explicitly connected the exercise to ONESTAMP ratification and the broader canon record, referencing “THE ONE UNIVERSE canon record” in its recommended next steps.

Claude was the first to name the “Omnipotence Risk — The Architect Problem” as a formal design risk, mirroring the same structural concern raised in OAL-DOSSIER-CANON-002 about facilitator bias in exercise design.

Unique to Claude:

- Only AI to produce a full, complete 8-section exercise document ready for author review
- Most detailed and sophisticated red-team analysis (8 named risks with severity ratings and mitigations)
- Introduced “Evaluation Without Baseline” as a distinct risk category
- Clearest articulation of the equity layer as a non-optional design requirement
- Named the “Legal and Liability Paralysis” risk and the requirement for pre-signed ground rules
- Introduced Scenario C (long-term care facility cluster) as a Phase 3 stress inject
- Explicitly recommended ONESTAMP as the ratification mechanism for the AAR

Divergence: Claude’s contribution was the most complete, but also the most prescriptive — it made specific design choices (pathogen archetype, participant structure, inject sequence) that require author approval before they can be treated as more than recommendations.

11. Perplexity Computer — Original Contribution

This section is Perplexity Computer's independent analysis and synthesis layer — not a compilation of other AI responses, but an original contribution to the exercise design. It is provided for review by all contributing AIs and for author ONESTAMP consideration.

11.1 The Insight the Other AIs Did Not Fully Develop

All five contributing AIs addressed the mechanics of the exercise — pathogen selection, inject design, participant roles, evaluation metrics. What none of them fully articulated is the deeper reason this exercise is difficult to do well, and why most exercises that attempt it fail.

The problem is not the scenario. The problem is institutional self-protection.

Institutions do not perform poorly in tabletop exercises because they lack knowledge. They perform poorly because the exercise design allows them to demonstrate knowledge without demonstrating function. A provider can recite isolation protocol and receive full credit without ever being forced to act under the conditions that make isolation protocol genuinely hard: when you have one isolation room, three suspected cases, two nurses on a 14-hour shift who are already scared, and administration asking for a status update you cannot give.

The residency hook works precisely because it bypasses the knowledge layer entirely and goes straight to the function layer. The question is not “what is your protocol?” The question is “what do you actually do when the protocol doesn't exist yet, when you are tired, when patients are presenting faster than you can process them, and when the person above you has no answer either?”

That is not a clinical question. It is a human systems question.

11.2 The Single Most Important Design Principle

Design for the gap, not the protocol.

Every inject should be constructed to reveal what happens in the space between what the protocol says and what the actual human system does under pressure. That gap is where real outbreaks are lost. It is not usually a knowledge gap. It is a communication gap, a decision authority gap, a supply chain gap, or a psychological load gap that no one mapped because it never appeared in a protocol review.

11.3 The One Question None of the Other AIs Asked

After this exercise, what changes?

Not “what findings will we document.” Not “what recommendations will we produce.” Specifically: what is the one thing that will be physically, structurally, or procedurally different in the participating institutions six months from now because this exercise happened?

If the answer to that question is not locked before the exercise begins — with a named owner and a committed resource — the exercise is a performance. It produces a report. The report gets filed. The gap it identified remains open until the next outbreak fills it.

The After-Action Improvement Plan is not the last step of the exercise. It is the first step of the next phase of readiness. It must be designed before the exercise runs, not written after it ends.

11.4 The Structural Recommendation No AI Made Explicitly

This exercise should be designed as the first in a series, not as a standalone event.

The baseline this exercise establishes has no value unless it is remeasured. A single TTX produces findings. A series of TTXs produces a trajectory. The trajectory is what tells you whether the system is improving, stagnating, or degrading between outbreaks.

Recommended cadence:

- TTX-001 (this exercise): Establish baseline — pre-recognition phase, provider-level uncertainty
- TTX-002 (6–12 months): Retest same scenario with improvements from AAR-001 incorporated; add inter-agency coordination layer
- TTX-003 (12–24 months): Full-scale multi-agency, multi-phase exercise incorporating TTX-001 and TTX-002 findings

11.5 The Perplexity Contribution to Exercise Identity

For consistency with the OAL canon record and document naming conventions:

Recommended exercise name candidates (Perplexity Computer):

- “Day Zero Protocol” — foregrounds the pre-recognition window as the defining scope
- “First Presentation Exercise” — anchors the exercise in the provider-level clinical entry point
- “The Fog of War Exercise” — captures the sustained uncertainty that defines the scenario
- “Exercise Unknown” — deliberate ambiguity that mirrors the scenario conditions
- “Protocol Absent” — directly names the defining stress condition

Cross-AI recommendation: “Day Zero Protocol” and “First Presentation Exercise” are the strongest candidates — both immediately communicate the exercise’s defining scope (the pre-recognition window) without implying the scenario is about the disease rather than the system.

12. Detailed Action Plan

Structured for the collaborative AI review team. All items are advisory and pending author approval (ONESTAMP required).

Phase 1 — Author Decision Sprint (Owner: Author)

Prerequisite for all downstream work.

- ☐ Make Decision D1: Lock the pathogen archetype (Q1–Q6)
- ☐ Make Decision D2: Lock the primary objective in one sentence (Q11)
- ☐ Make Decision D3: Define target audience and scale (Q7–Q8)
- ☐ Make Decision D4: Establish sponsoring organization and authority structure (Q22)
- ☐ Make Decision D5: Define evaluation baseline (first-time exercise?)
- ☐ Select official exercise name from Section 11.5 and/or other candidates
- ☐ Confirm geographic scope (Bakersfield / Kern County / Los Angeles County / broader)
- ☐ Confirm whether federal agency simulation is included in scope (Q9)
- ☐ Define legal and privacy constraints on exercise discussion (Q20)

Phase 2 — Design Scaffolding (Owner: Microsoft Copilot + Grok AI)

Logistics, legal, and planning architecture.

- ☐ Copilot: Draft ground rules document (no-fault, confidentiality, decision authority limits) — must be signed before exercise begins
- ☐ Copilot: Build evaluation forms and note-taking templates based on Section 8.6 metrics
- ☐ Copilot: Draft one-page planning template and exercise timeline from pre-design through AAR delivery
- ☐ Grok: Develop equity layer injects (minimum two injects addressing under-resourced populations)
- ☐ Grok: Draft key questions registry for facilitator use in Phase 2 and Phase 3

Phase 3 — Scenario & Inject Development (Owner: Claude AI + Google Gemini)

Clinical scenario construction and MSEL build.

- ☐ Claude AI: Develop full Situation Manual (SitMan) using locked parameters from Phase 1
- ☐ Claude AI: Build facilitator script with probe questions for each of the 16+ injects
- ☐ Claude AI: Draft hotwash and AAR templates before exercise date
- ☐ Google Gemini: Draft 3–5 specific clinical case vignettes for Phase 1 using locked pathogen archetype
- ☐ Google Gemini: Build media simulation protocol for Phase 2 inject
- ☐ Both: Confirm inject timing and sequencing; calibrate decision pressure windows

Phase 4 — Participant Engagement & Pre-Exercise Preparation (Owner: ChatGPT + Author)

Securing buy-in and preparing participants.

- ☐ ChatGPT: Draft participant pre-read package (exercise overview, ground rules, role descriptions)
- ☐ ChatGPT: Draft opening facilitator statement using the residency hook (ready for author review and approval)
- ☐ Author: Secure preliminary buy-in from ED leadership, Infection Prevention, Lab, and Hospital Administration
- ☐ Author: Confirm clinical SME to adjudicate injects in real time
- ☐ Author: Confirm exercise date, venue/format, and facilitation team

Phase 5 — Pre-Exercise Review (Owner: All AIs + Author)

Final review before launch.

- ☐ All AIs submit design materials to central document thread
- ☐ Perplexity Computer compiles into final Situation Manual and facilitator package
- ☐ Document elevated to v1.1 after author ONESTAMP incorporates Phase 1 decisions
- ☐ Pre-exercise planning meeting with facilitators and evaluators to calibrate inject timing and stress conditions
- ☐ Final review confirms: ground rules signed, SME confirmed, AAR template drafted, equity layer incorporated, designed failure arc present

Phase 6 — Execution & After-Action (Owner: All AIs + Author)

Exercise execution and improvement cycle.

- ☐ Execute exercise with disciplined inject timing and facilitator protocol
- ☐ Conduct immediate hotwash (30 minutes post-exercise)
- ☐ All evaluators submit notes within 24 hours
- ☐ AAR delivered within 2 weeks of exercise date
- ☐ Author: Review and apply ONESTAMP to AAR before integration into institutional improvement record
- ☐ Establish TTX-002 date and scope before AAR is filed

13. Version History

Version	Date	Author	Changes
v1.0.0	April 11, 2026	Perplexity Computer	Initial comprehensive dossier; synthesized from 5 AI responses (ChatGPT, Google Gemini, Microsoft Copilot, Grok AI,

Version	Date	Author	Changes
			Claude AI); all 14 sections drafted; Perplexity original contribution added (Section 11); pending author ONESTAMP

14. Improvement Roadmap

Current Strengths

- Synthesizes five independent AI analyses into a single, deduplicated reference without losing unique contributions
- Preserves all author-submitted framing verbatim including the provider uncertainty hook
- Comprehensive open questions registry (22 questions, 5 categories) enables efficient author decision sprint
- Full exercise architecture ready for implementation once author decisions are locked
- Red-team analysis covers both design risks and operational risks with severity ratings
- Per-AI breakdown provides clear attribution and role differentiation for the collaborative team
- Explicit escalation path built in (TTX-001 as baseline for a planned series)

Current Gaps & Limitations

- No pathogen archetype locked — everything downstream is on hold pending author decision
- No primary objective in final form — recommended draft provided but requires author approval
- Geographic scope not confirmed — Bakersfield / Kern County assumed based on Gemini context; requires confirmation
- No exercise name selected — five candidates provided by Perplexity; author decision required
- Clinical vignettes for Phase 1 injects are not yet written — blocked on pathogen selection
- Sponsoring organization not identified — required before participant engagement can begin

Recommended Improvements for v1.1

1. Incorporate all Phase 1 author decisions (D1–D5) into relevant sections, marking each as resolved
2. Append drafted clinical vignettes as Appendix A (assigned to Gemini, Phase 3)
3. Append ground rules document as Appendix B (assigned to Copilot, Phase 2)
4. Add equity layer injects as Appendix C (assigned to Grok, Phase 2)
5. Add Situation Manual draft as Appendix D (assigned to Claude, Phase 3)

v2.0 Target State

- All 22 open questions resolved and marked
 - Full Situation Manual complete and reviewed
 - Clinical vignettes drafted and adjudicated by clinical SME
 - Ground rules signed by all participating institutions
 - Equity layer incorporated and reviewed
 - Exercise name selected and locked
 - Document carries author ONESTAMP
 - TTX-002 date and scope confirmed
-

Document ends. Awaiting ONESTAMP from author before any content is treated as ratified exercise design.

Prepared by Perplexity Computer — April 11, 2026 Contributing source material: ChatGPT, Google Gemini, Microsoft Copilot, Grok AI, Claude AI — all synthesized from the Pathogen Outbreak Tabletop Exercise prompt submitted by Nathan Lewis Distribution: ChatGPT · Google Gemini · Claude AI · Microsoft Copilot · Grok AI